

Rule-Based AI Chatbot ■

A simple Rule-Based AI Chatbot developed in Python.

■ Project Overview

This project is a simple chatbot that interacts with users through the terminal. It gives predefined responses based on the user's input. The chatbot uses Python conditional statements such as if, elif, and else to understand specific commands.

■ Features

- Responds to "hello" and "hi"
- Responds to "how are you?"
- Tells its name when asked
- Exits when the user types "bye"
- Gives a default response for unknown questions
- Converts user input to lowercase

■■ Technologies Used

- Python 3
- if / elif / else statements
- while loop
- input() function

■ How to Run

1. Install Python 3 on your computer.
2. Open the project in VS Code.
3. Open the terminal.
4. Run the following command:

```
python chatbot.py
```

■ Example Conversation

■ Welcome to Rule-Based AI Chatbot!
Type 'bye' to exit.

You: hello

Bot: Hello! How can I help you?

You: how are you?

Bot: I'm fine. Thank you!

You: what is your name?

Bot: My name is DecodeBot.

You: bye

Bot: Goodbye! Have a nice day.

■ How It Works

The chatbot continuously takes input from the user using a while True loop. The user's input is converted into lowercase. The chatbot then checks the input using conditional statements and gives the appropriate response. If the user types bye, the chatbot says goodbye and stops the program using break. For an unknown input, it responds: "Bot: Sorry, I don't understand."

■ Purpose

The purpose of this project is to understand the basic concepts of Python programming, conditional statements, loops, user input, basic chatbot logic, and Rule-Based Artificial Intelligence.

■ Future Improvements

- More questions and responses
- Multiple responses for the same question
- GUI interface
- Voice input and output
- Natural Language Processing (NLP)
- Machine Learning
- Conversation history

■■■ Author

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■ This is a beginner-friendly Rule-Based AI Chatbot built using Python.